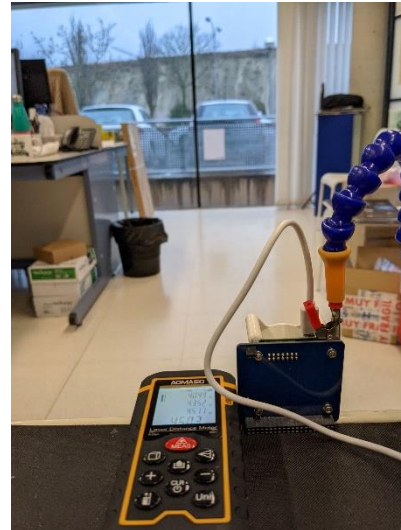


High Accuracy Level Sensing Test

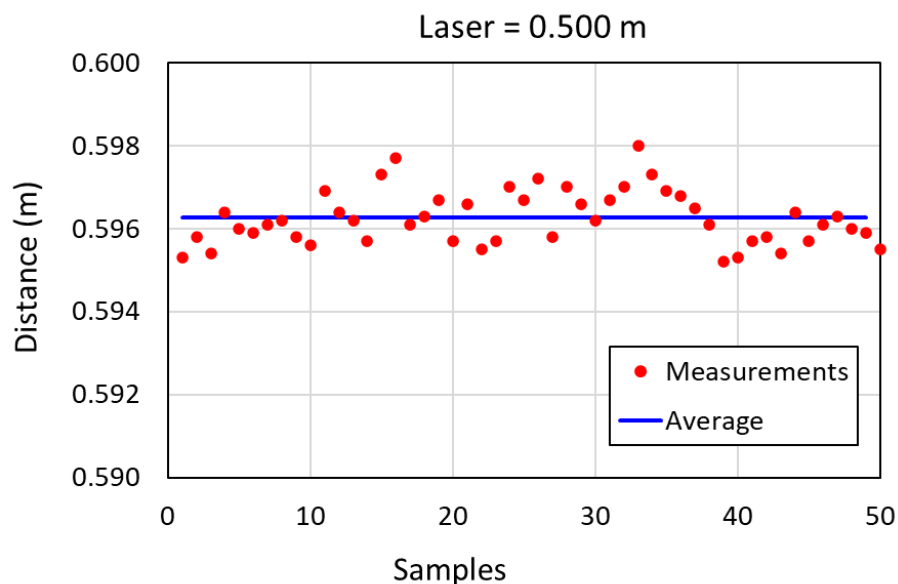
Test 1

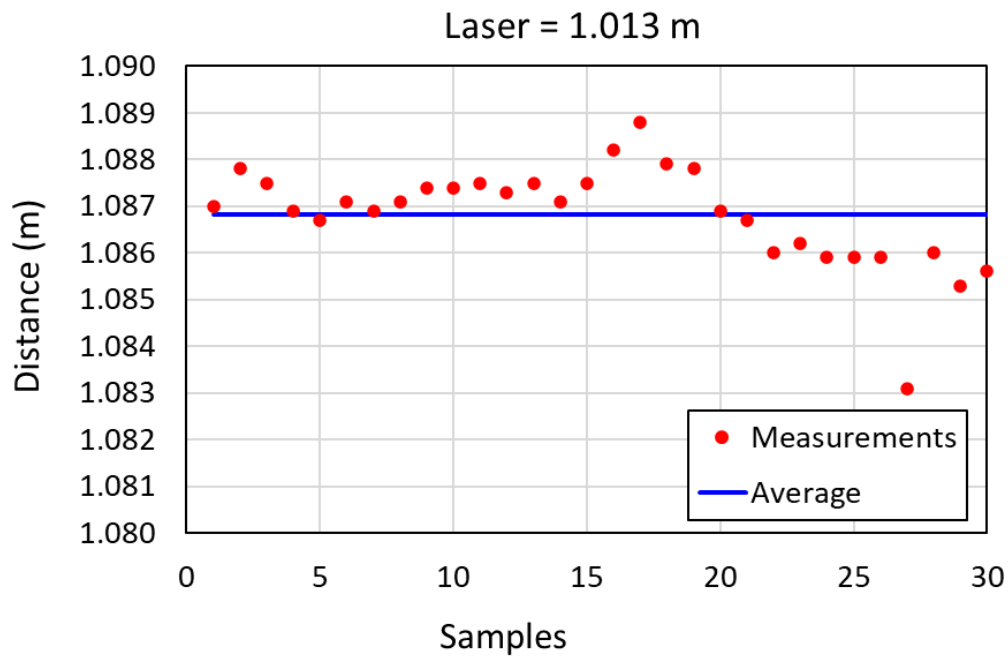
Comparison between laser and radar in an indoor scenario with the radar connected to a laptop.



Configuration:

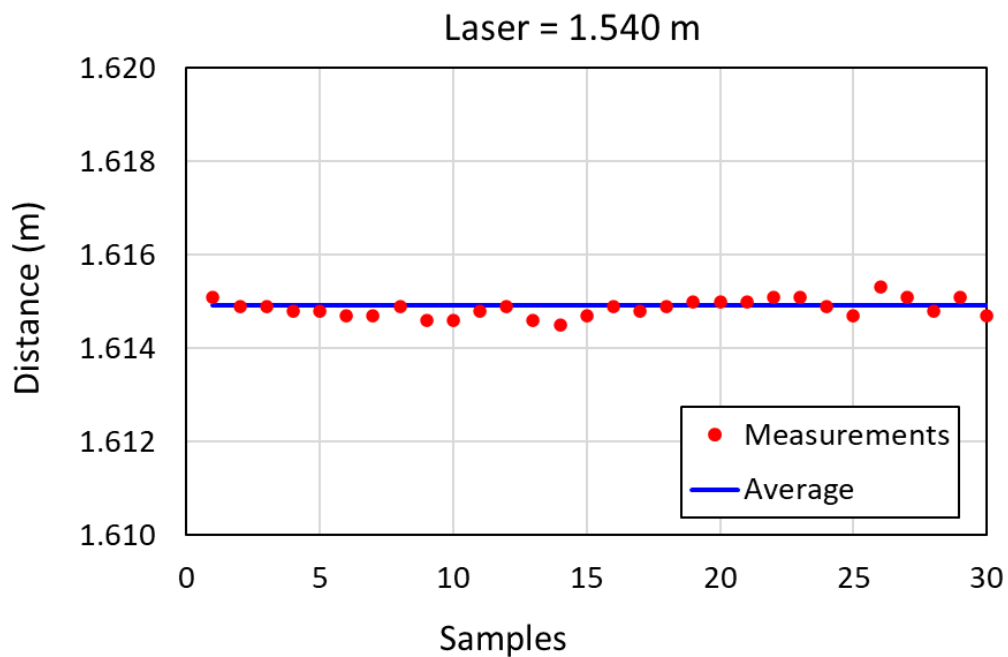
- Sample rate = 0.5 seg
- Maximum distance = 44 meters.





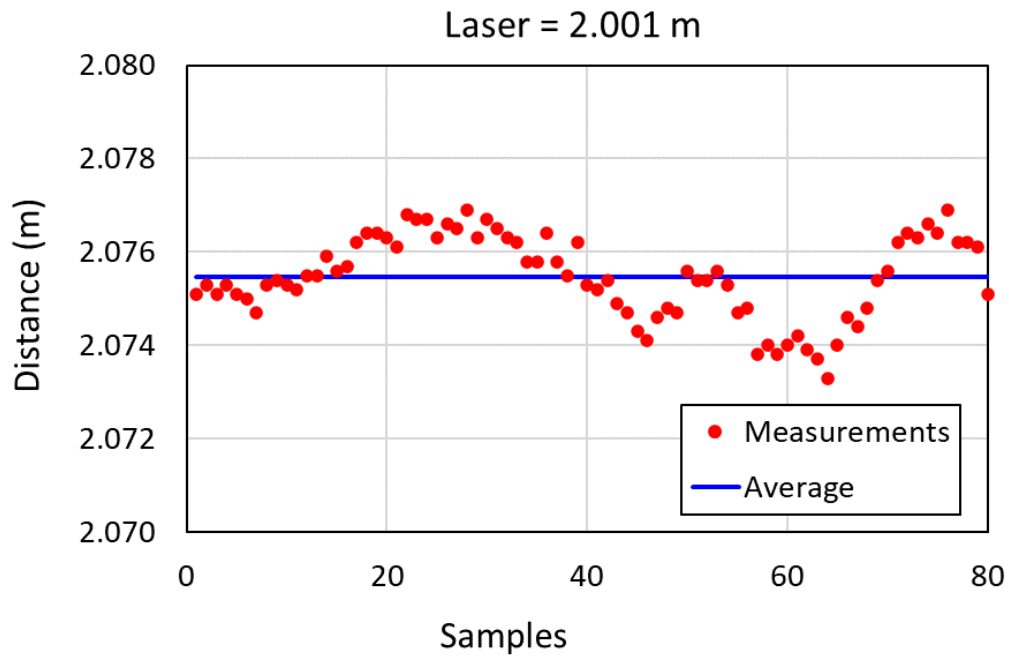
$$\text{Error} = 1.013 - 1.087 = -0.074 \text{ m}$$

$$\text{Deviation} = \pm 3 \text{ mm}$$



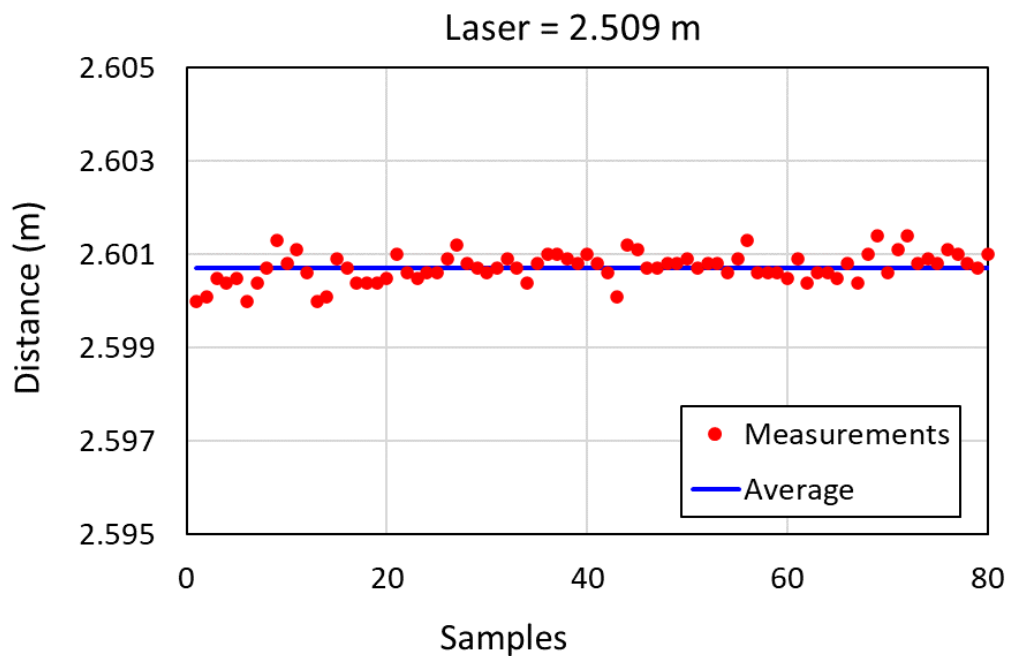
$$\text{Error} = 1.540 - 1.615 = -0.075 \text{ m}$$

$$\text{Deviation} = \pm 1 \text{ mm}$$



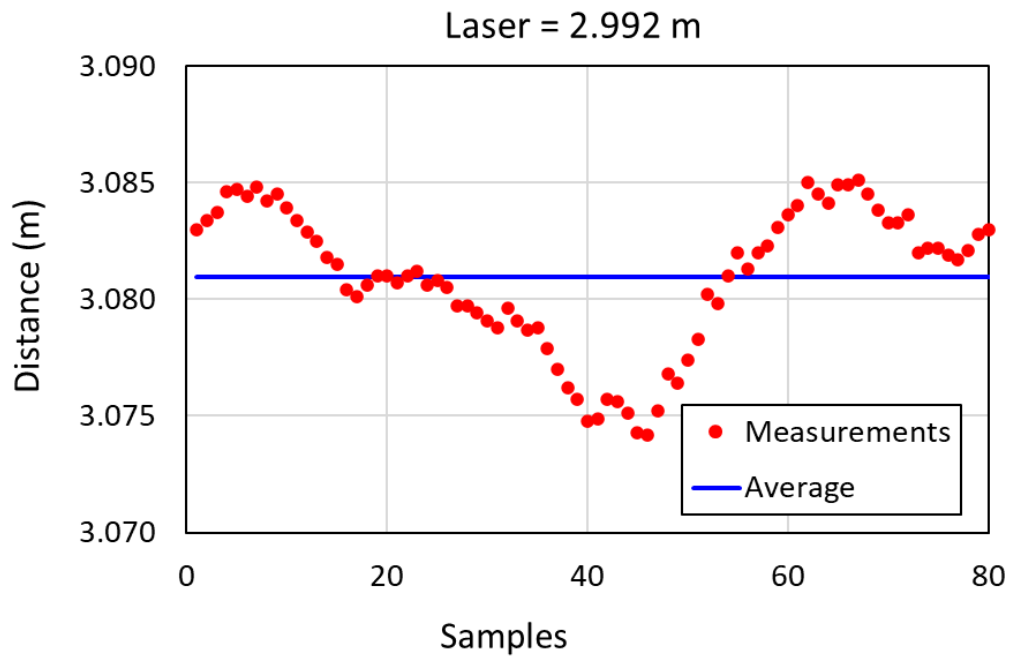
$$\text{Error} = 2.001 - 2.075 = -0.074 \text{ m}$$

$$\text{Deviation} = \pm 2 \text{ mm}$$



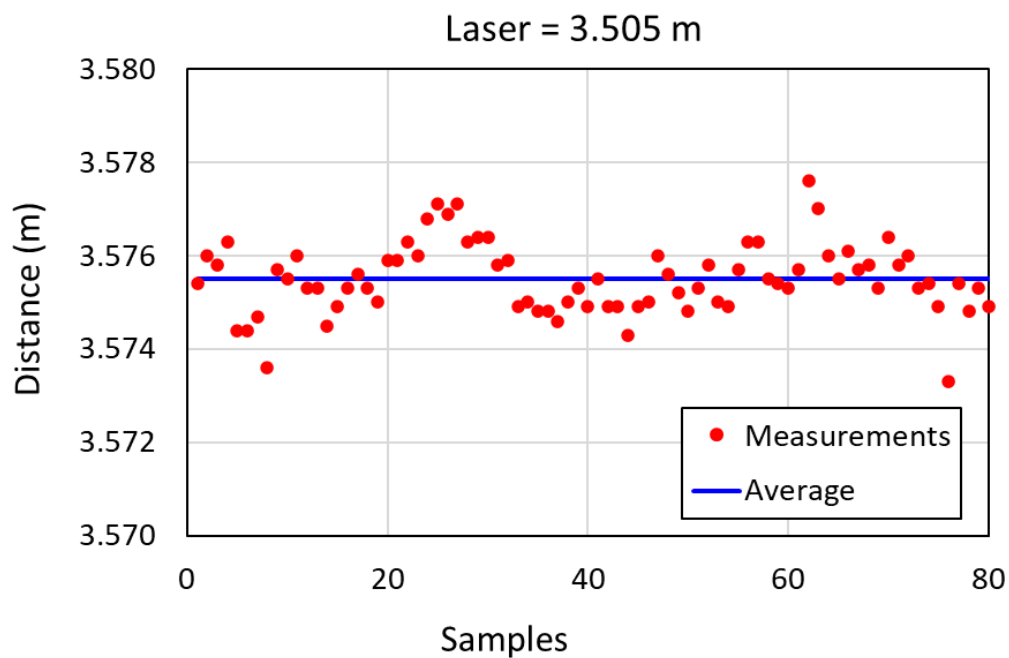
$$\text{Error} = 2.509 - 2.601 = -0.092 \text{ m}$$

$$\text{Deviation} = \pm 0.5 \text{ mm}$$



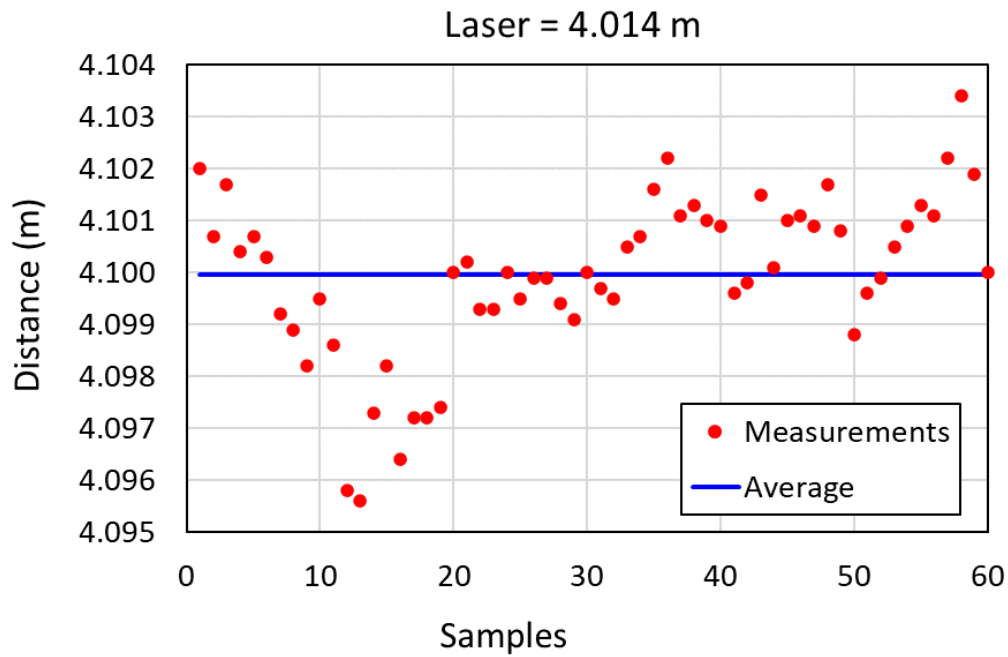
$$\text{Error} = 2.992 - 3.081 = -0.089 \text{ m}$$

$$\text{Deviation} = \pm 5 \text{ mm}$$



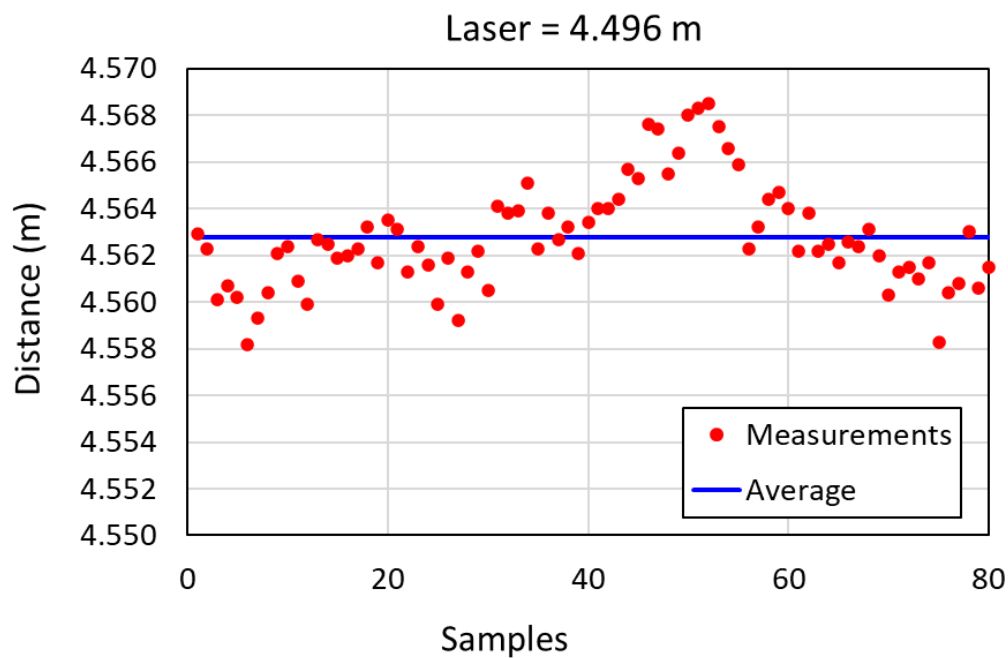
$$\text{Error} = 3.505 - 3.575 = -0.070 \text{ m}$$

$$\text{Deviation} = \pm 2 \text{ mm}$$



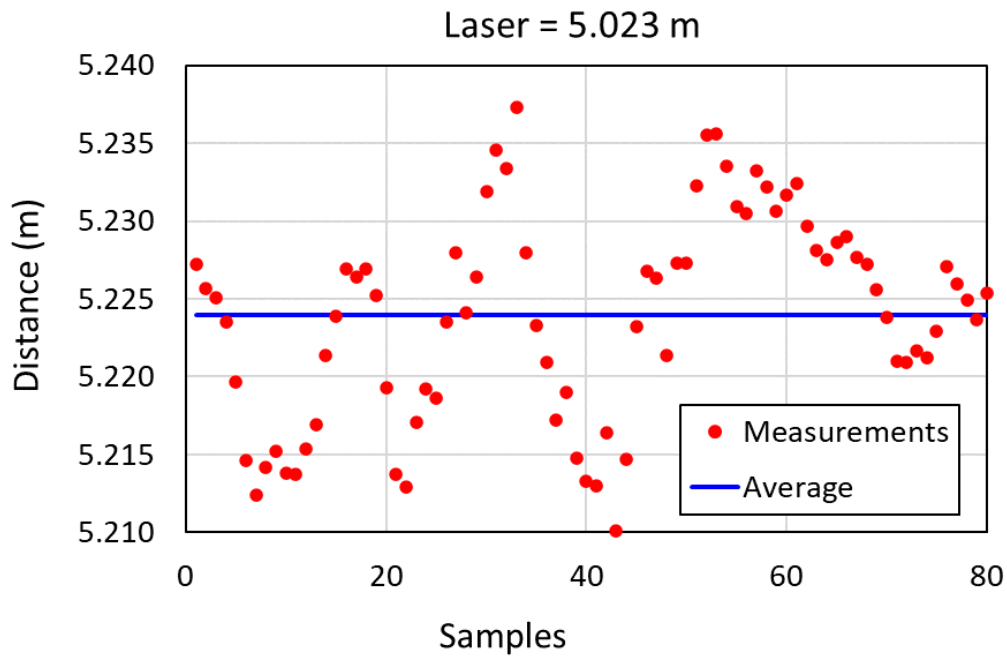
$$\text{Error} = 4.014 - 4.100 = -0.086 \text{ m}$$

$$\text{Deviation} = \pm 4 \text{ mm}$$



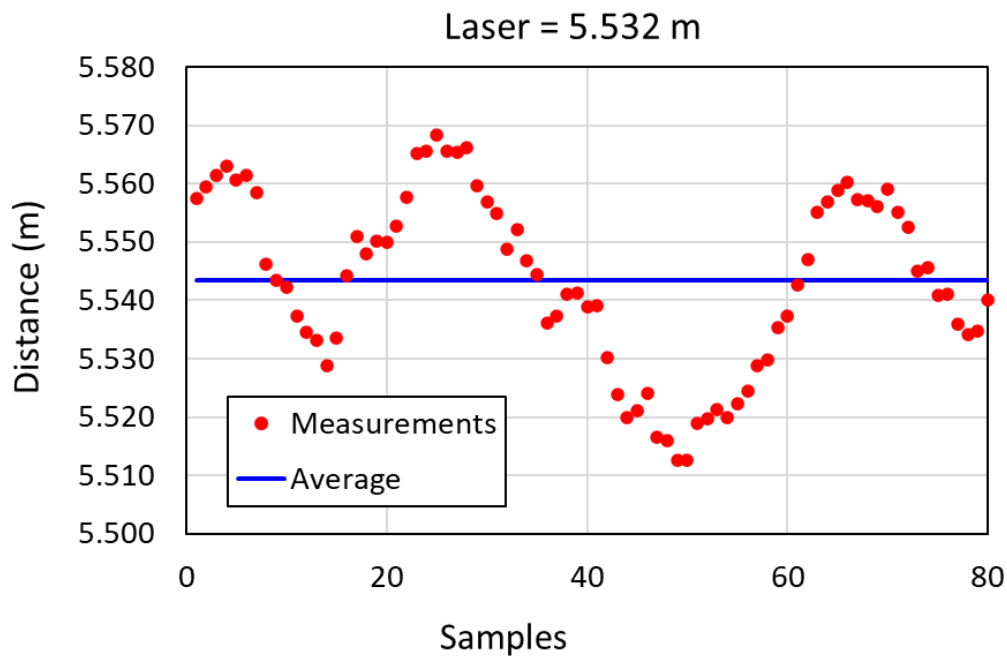
$$\text{Error} = 4.496 - 4.563 = -0.067 \text{ m}$$

$$\text{Deviation} = \pm 5 \text{ mm}$$



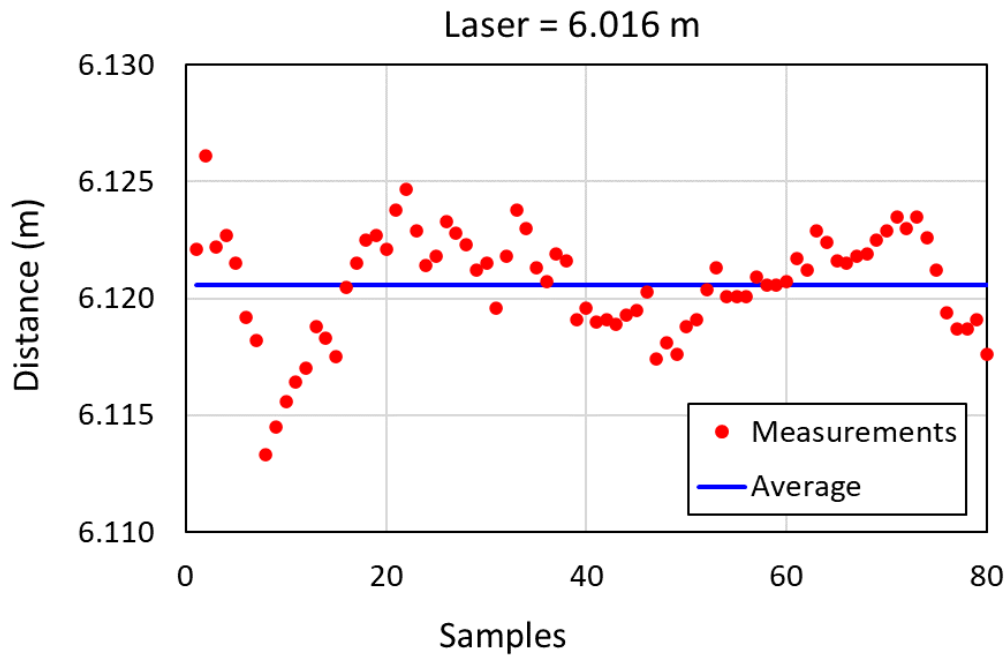
$$\text{Error} = 5.023 - 5.224 = -0.201 \text{ m}$$

$$\text{Deviation} = \pm 13.5 \text{ mm}$$



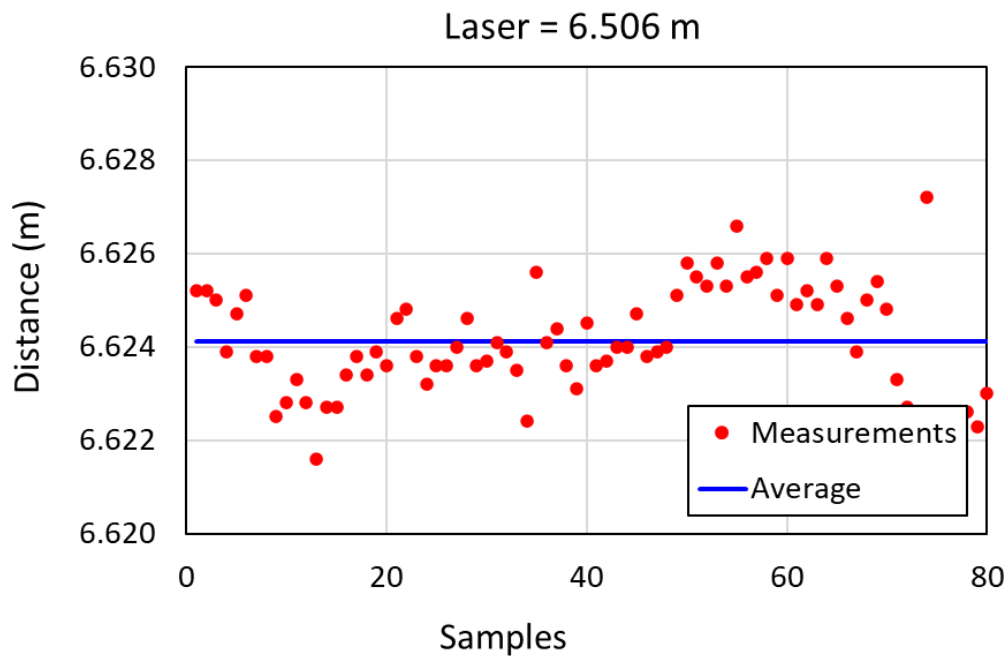
$$\text{Error} = 5.532 - 5.543 = -0.011 \text{ m}$$

$$\text{Deviation} = \pm 30 \text{ mm}$$



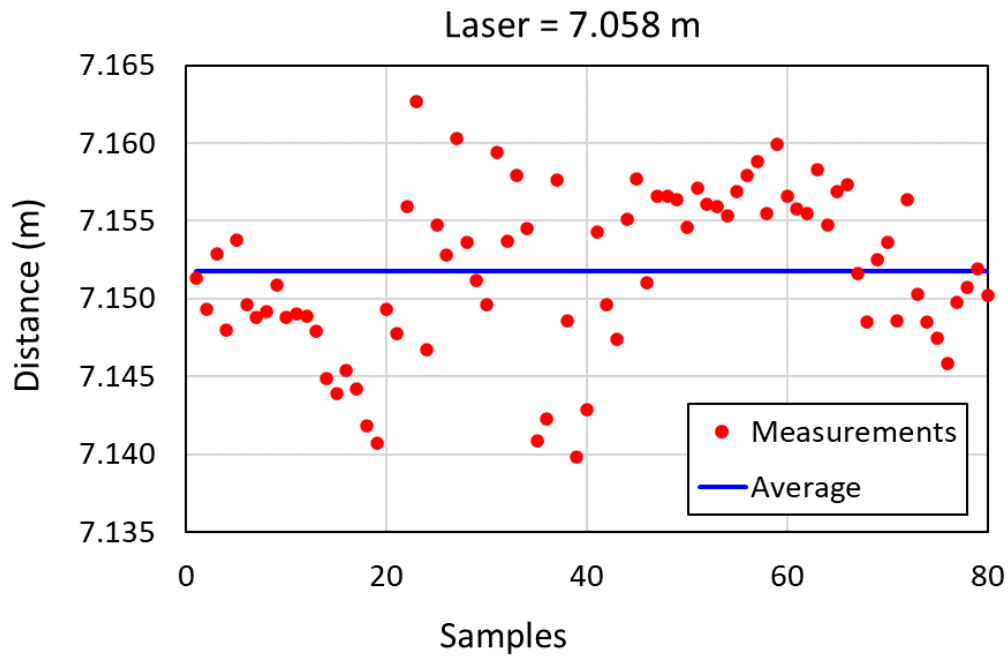
$$\text{Error} = 6.016 - 6.121 = -0.105 \text{ m}$$

$$\text{Deviation} = \pm 6.5 \text{ mm}$$



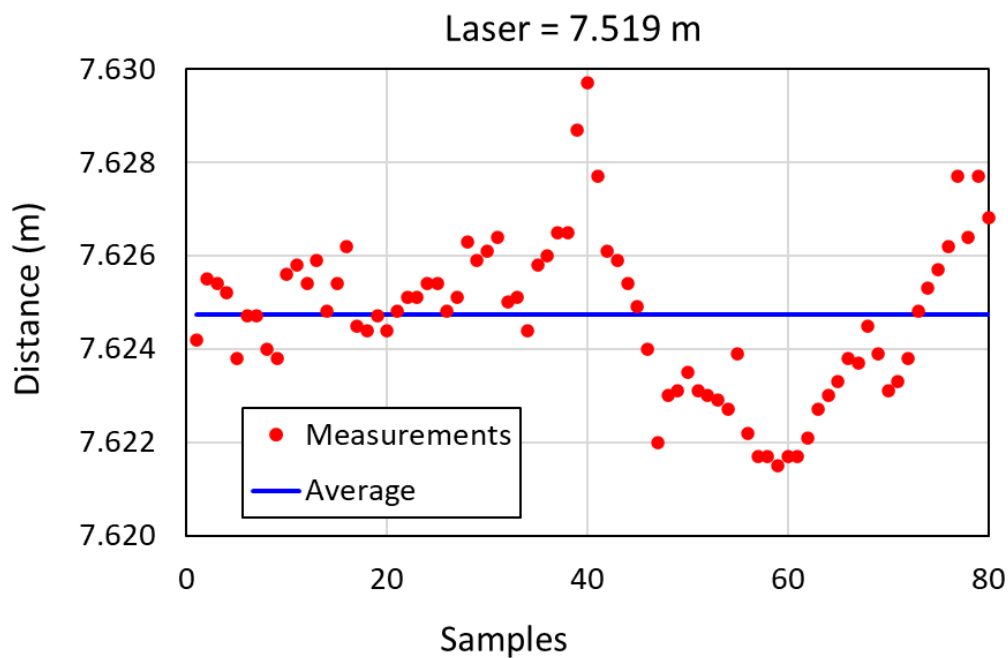
$$\text{Error} = 6.506 - 6.624 = -0.118 \text{ m}$$

$$\text{Deviation} = \pm 2.5 \text{ mm}$$



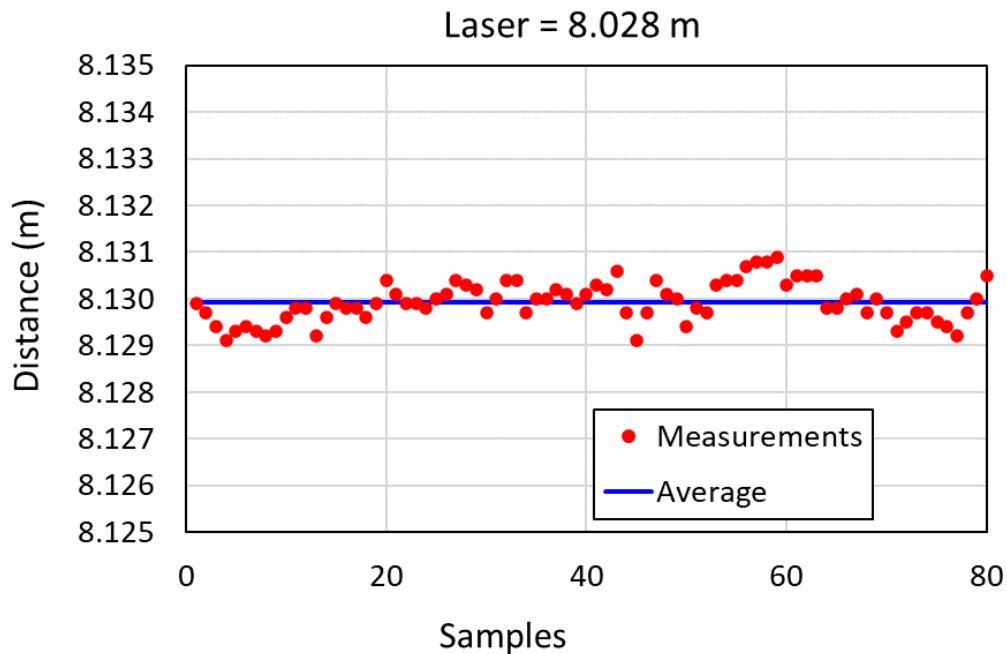
$$\text{Error} = 7.058 - 7.152 = -0.094 \text{ m}$$

$$\text{Deviation} = \pm 12 \text{ mm}$$



$$\text{Error} = 7.519 - 7.625 = -0.106 \text{ m}$$

$$\text{Deviation} = \pm 4 \text{ mm}$$



$$\text{Error} = 8.028 - 8.130 = -0.102 \text{ m}$$

$$\text{Deviation} = \pm 1 \text{ mm}$$

Conclusions:

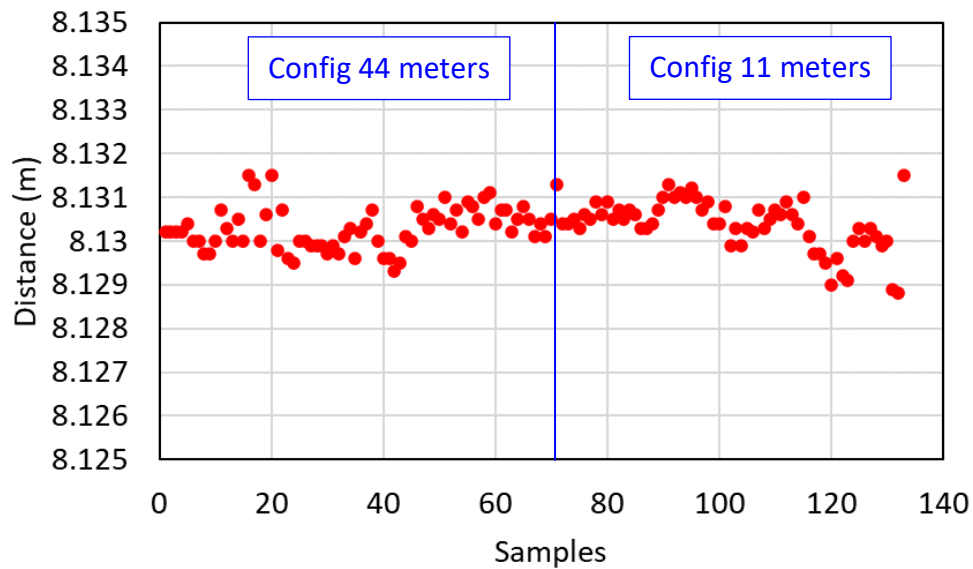
Under the conditions of an indoor scenario and up to 8 meters:

- There is a constant offset of around 0.1 meters, which does not depend on distance.
- The deviation in the measurements is very low and does not depend on distance.
- In an indoor scenario, there are multiple reflections with surrounding objects. Although the lens collimates the beam to a focus, there is always power outside of the beam that influences the measurement slightly. For example, a person moving near (or not so near) the measurement zone will slightly affect the measurement. This is why the deviation is higher at some distances than at others; it depends on the particular scenario at the moment of the experiment.

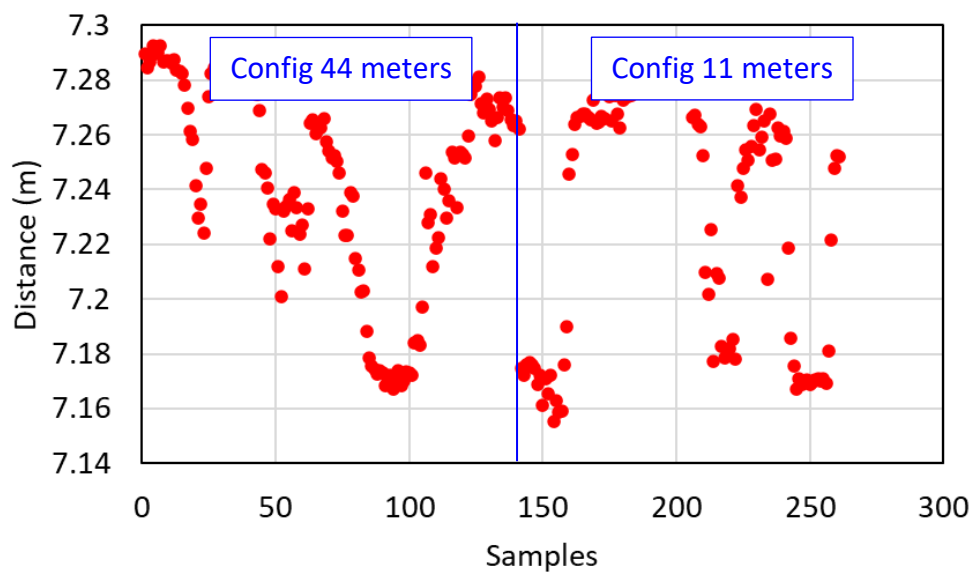
Test 2

Comparison between two configurations: with maximum distance 44 meters (smaller BW) and 11 meters (bigger BW).

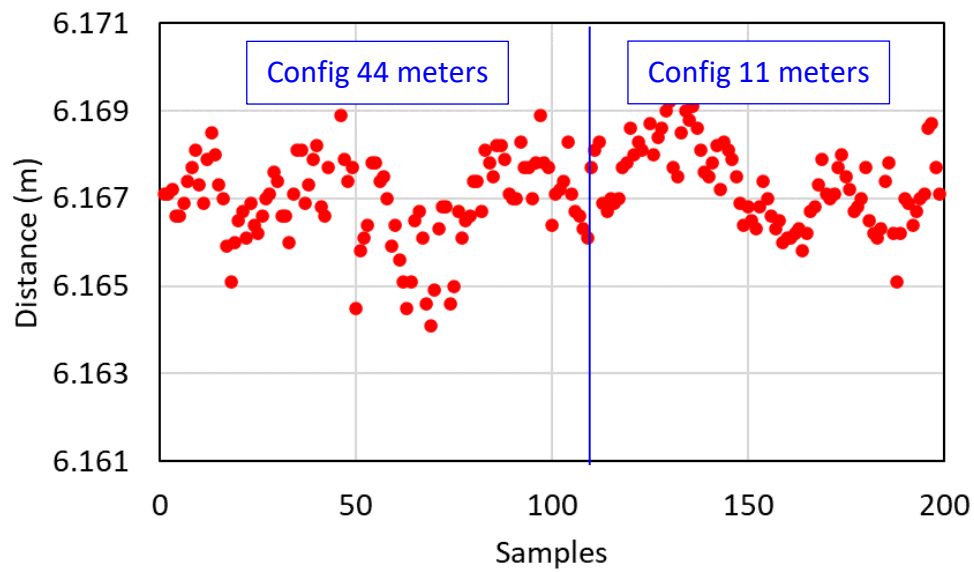
Laser = 8.027



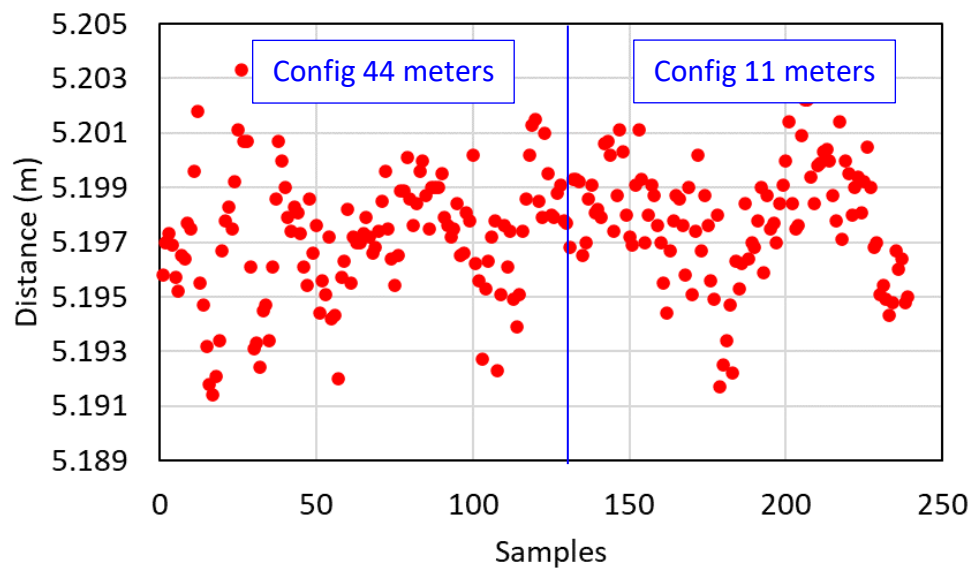
Laser = 7.074



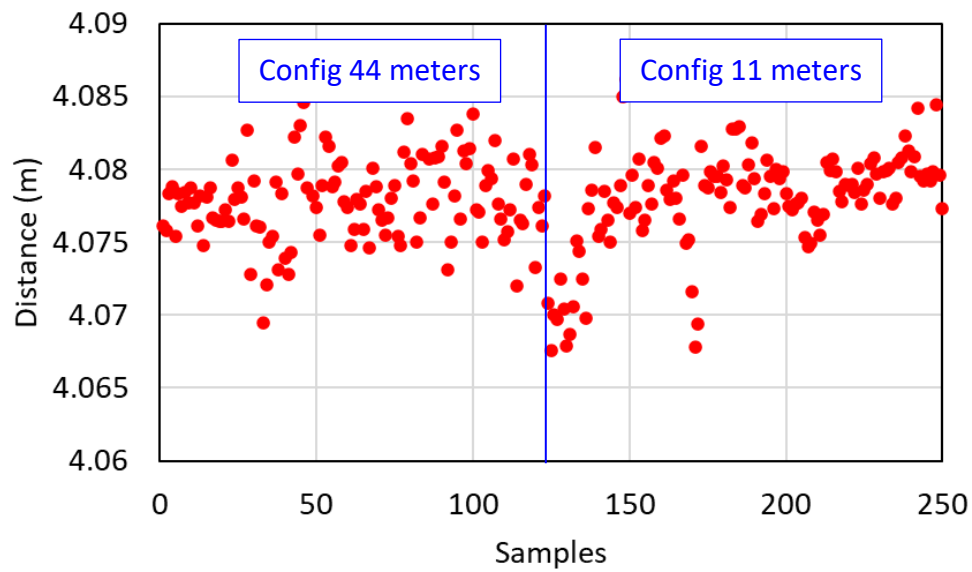
Laser = 6.052



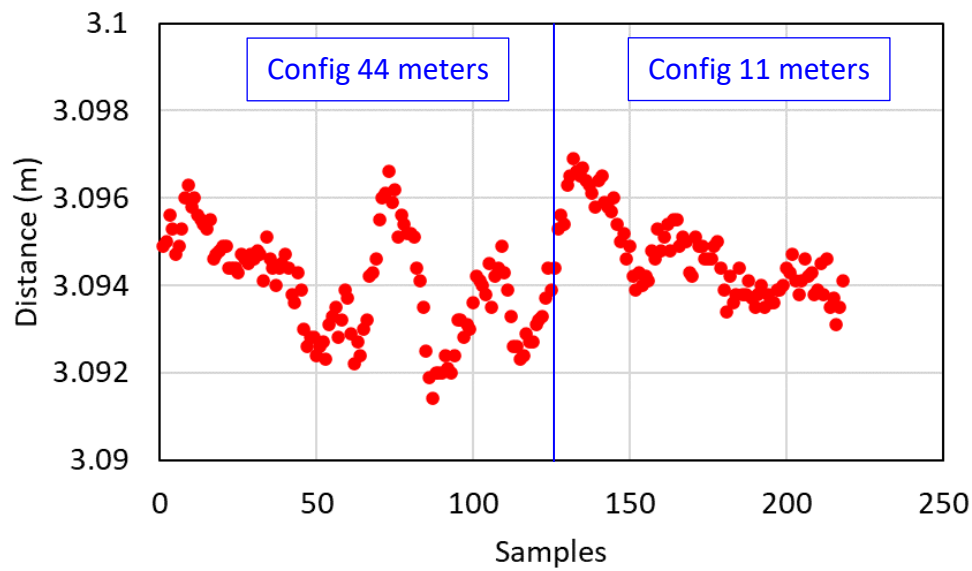
Laser = 5.054

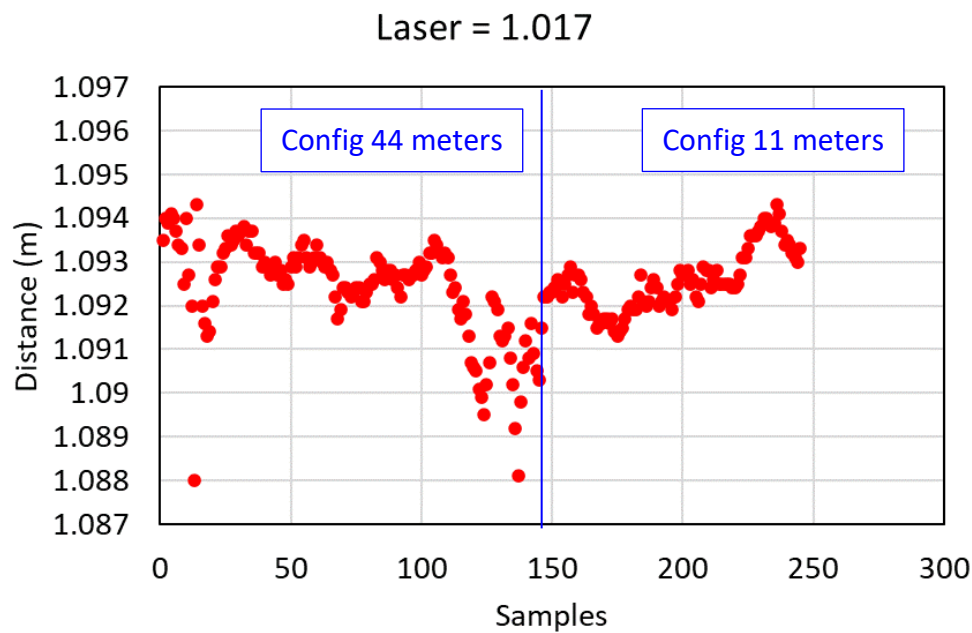
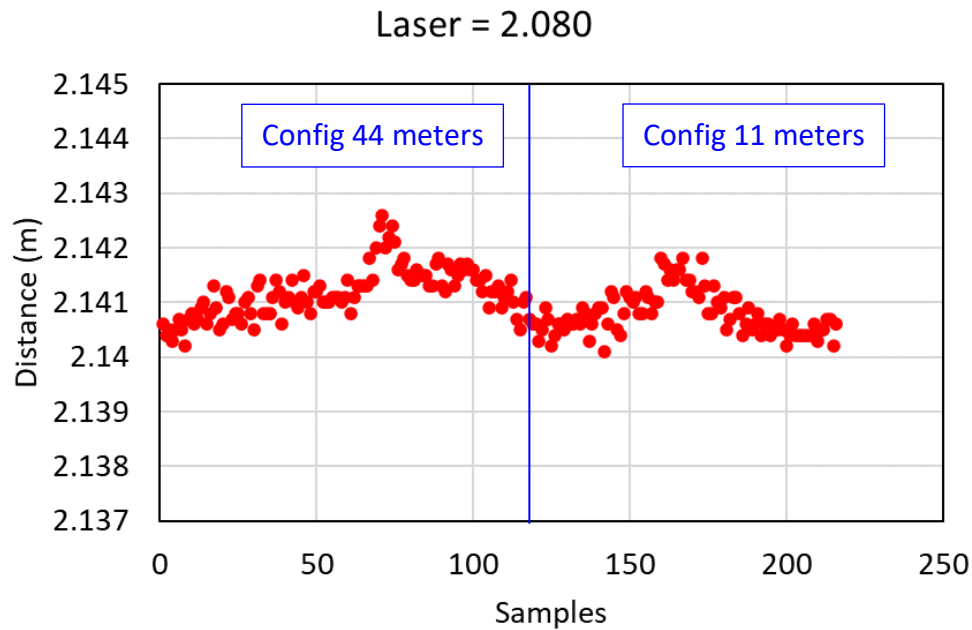


Laser = 4.018



Laser = 3.015



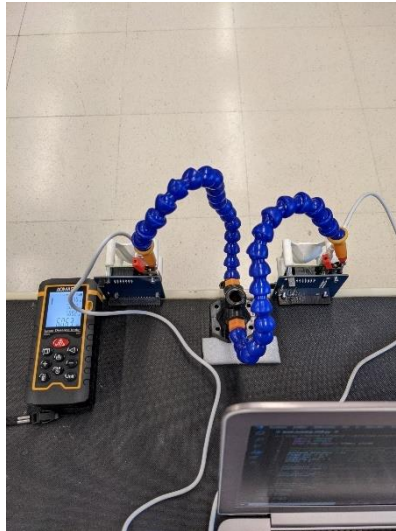


Conclusions

Configuration (whether bigger or smaller bandwidth) does not influence in the accuracy, at least up to 8 meters.

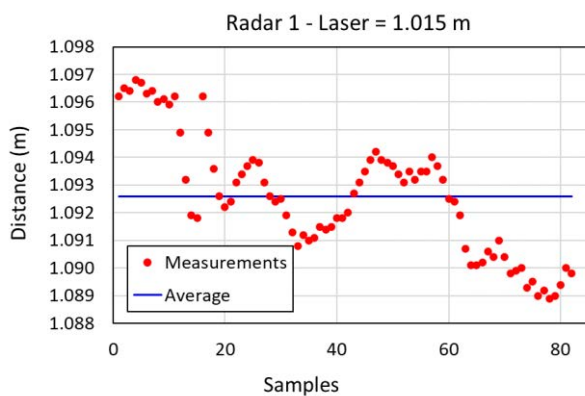
Test 3

Comparison between two radars and a laser. One radar is new and the other one has been heavily used.



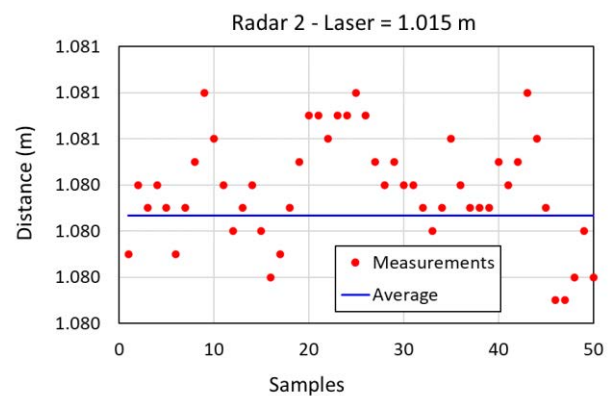
Configuration:

- Sample rate = 0.5 seg
- Maximum distance = 44 meters.



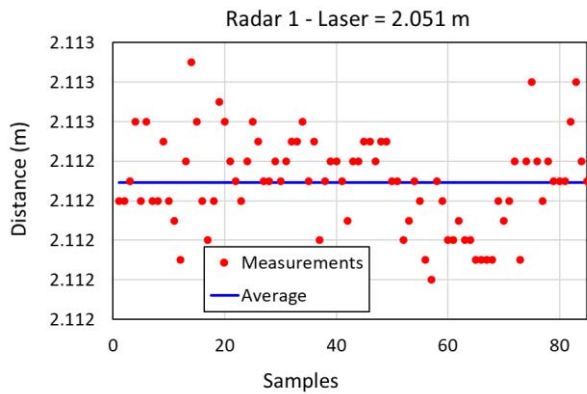
$$\text{Error Radar1} = 1.015 - 1.080 = -0.065 \text{ m}$$

$$\text{Deviation Radar1} = \pm 4 \text{ mm}$$



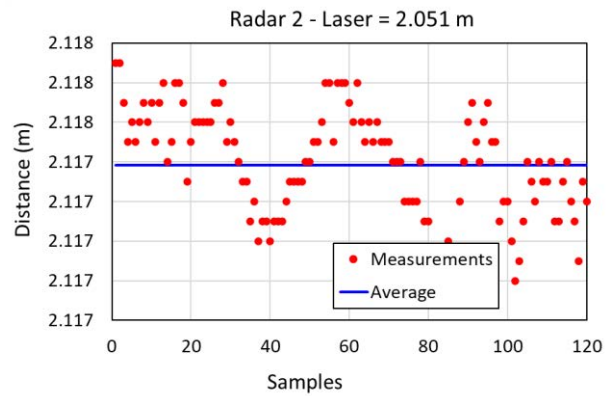
$$\text{Error Radar2} = 1.015 - 1.080 = -0.065 \text{ m}$$

$$\text{Deviation Radar2} = \pm 0.5 \text{ mm}$$



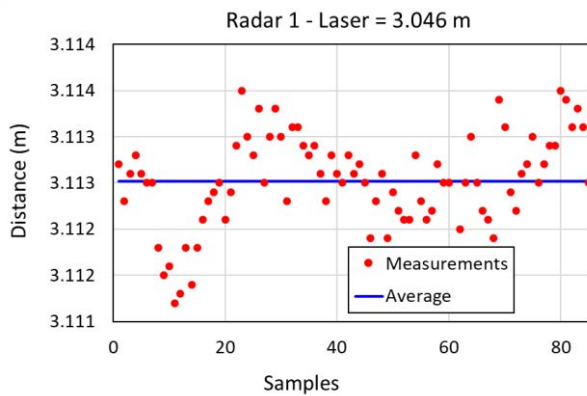
Error Radar1 = $2.051 - 2.112 = -0.061$ m

Deviation Radar1 = ± 0.5 mm



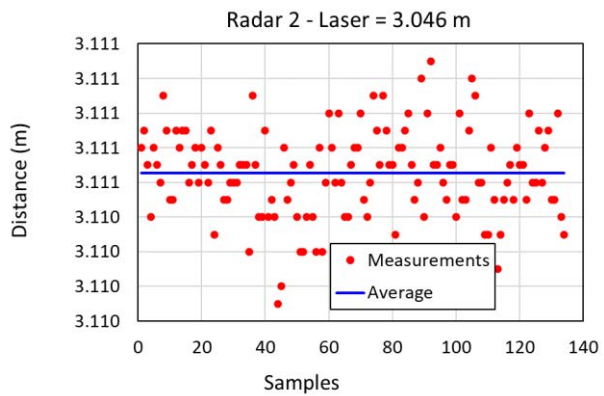
Error Radar2 = $2.051 - 2.117 = -0.066$ m

Deviation Radar2 = ± 0.5 mm



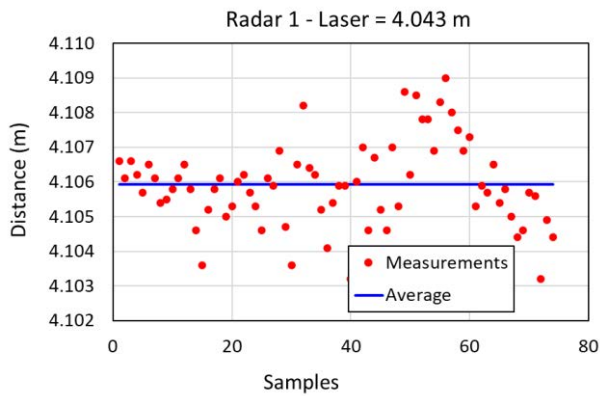
Error Radar1 = $3.046 - 3.113 = -0.067$ m

Deviation Radar1 = ± 1.5 mm



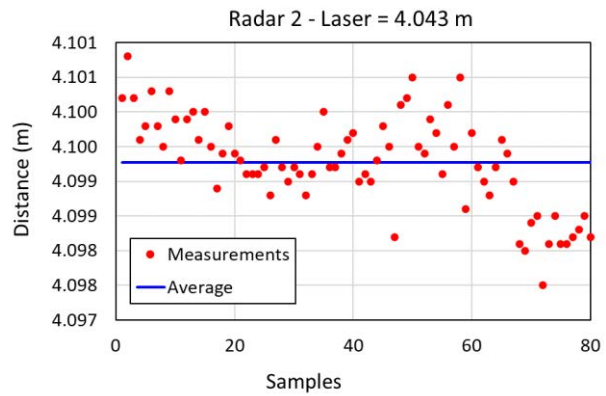
Error Radar2 = $3.046 - 3.111 = -0.065$ m

Deviation Radar2 = ± 0.5 mm



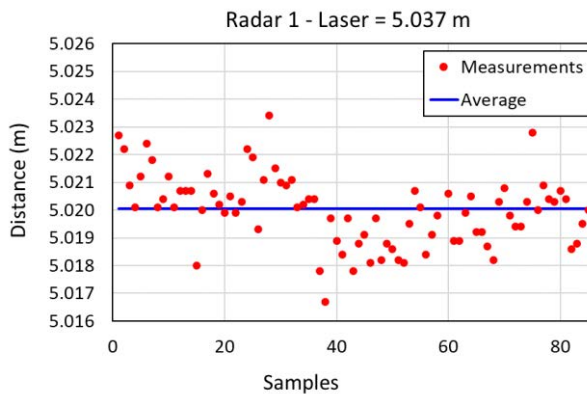
$$\text{Error Radar1} = 4.043 - 4.106 = -0.063 \text{ m}$$

$$\text{Deviation Radar1} = \pm 3 \text{ mm}$$



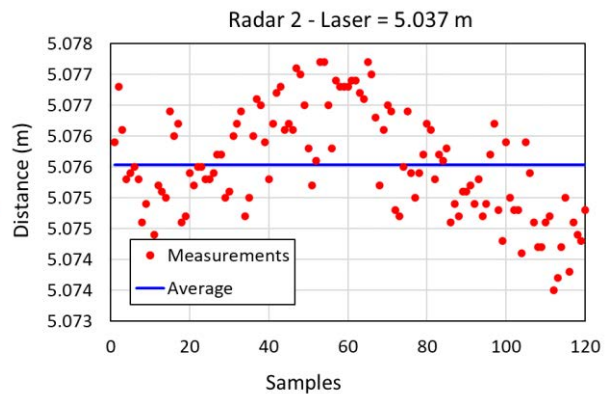
$$\text{Error Radar2} = 4.043 - 4.099 = -0.056 \text{ m}$$

$$\text{Deviation Radar2} = \pm 1.5 \text{ mm}$$



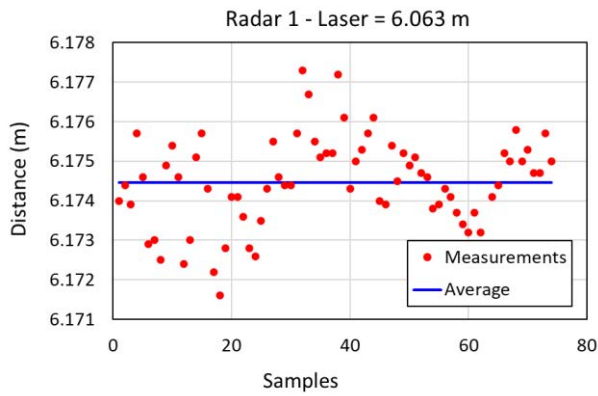
$$\text{Error Radar1} = 5.037 - 5.020 = -0.017 \text{ m}$$

$$\text{Deviation Radar1} = \pm 3.5 \text{ mm}$$



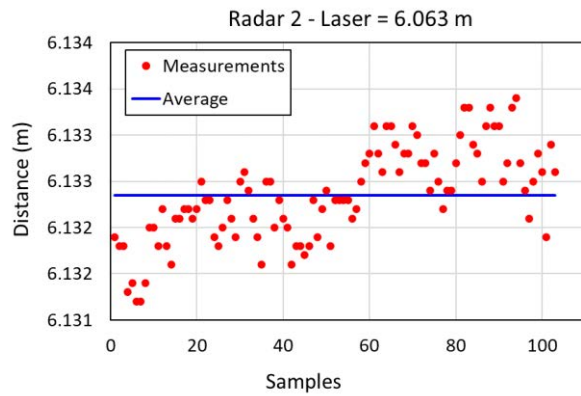
$$\text{Error Radar2} = 5.037 - 5.076 = -0.039 \text{ m}$$

$$\text{Deviation Radar2} = \pm 0.5 \text{ mm}$$



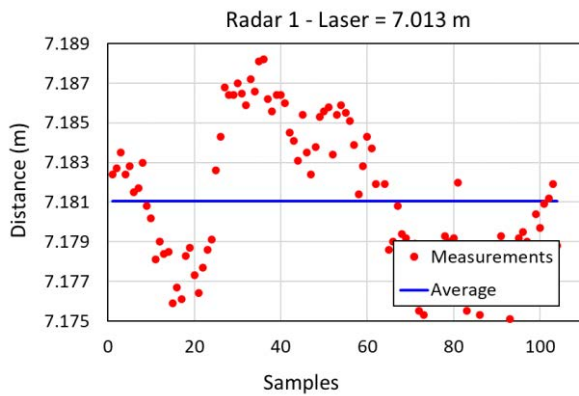
$$\text{Error Radar1} = 6.063 - 6.174 = -0.111 \text{ m}$$

$$\text{Deviation Radar1} = \pm 4 \text{ mm}$$



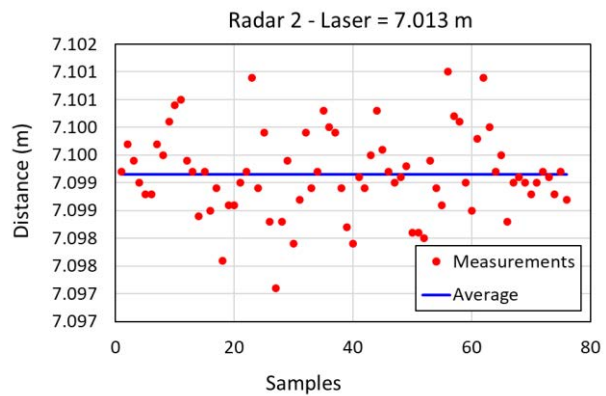
$$\text{Error Radar2} = 6.063 - 6.132 = -0.069 \text{ m}$$

$$\text{Deviation Radar2} = \pm 1.5 \text{ mm}$$



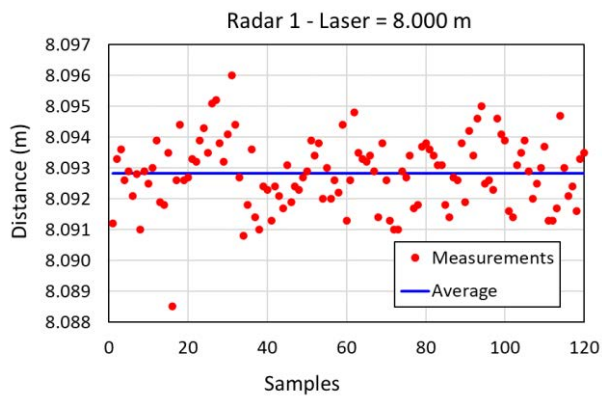
$$\text{Error Radar1} = 7.013 - 7.181 = -0.168 \text{ m}$$

$$\text{Deviation Radar1} = \pm 4 \text{ mm}$$



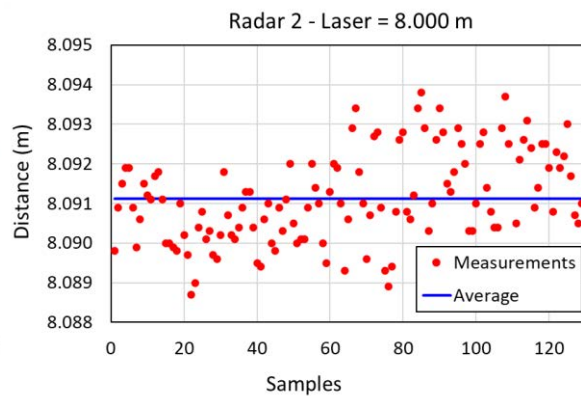
$$\text{Error Radar2} = 7.013 - 7.099 = -0.086 \text{ m}$$

$$\text{Deviation Radar2} = \pm 2 \text{ mm}$$



$$\text{Error Radar1} = 8.000 - 8.093 = -0.093 \text{ m}$$

$$\text{Deviation Radar1} = \pm 2.5 \text{ mm}$$



$$\text{Error Radar2} = 8.000 - 8.091 = -0.091 \text{ m}$$

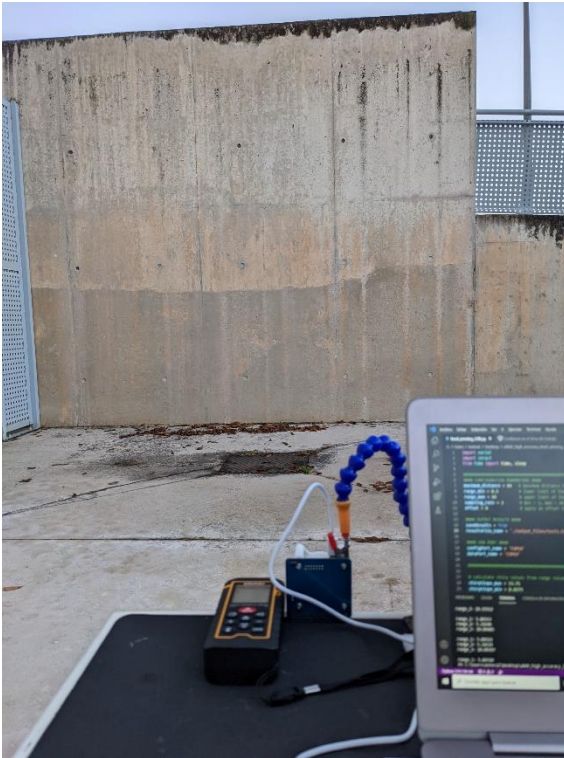
$$\text{Deviation Radar2} = \pm 2.5 \text{ mm}$$

Conclusions

- Similar performance to Test 1
- There are not significant differences between radars.

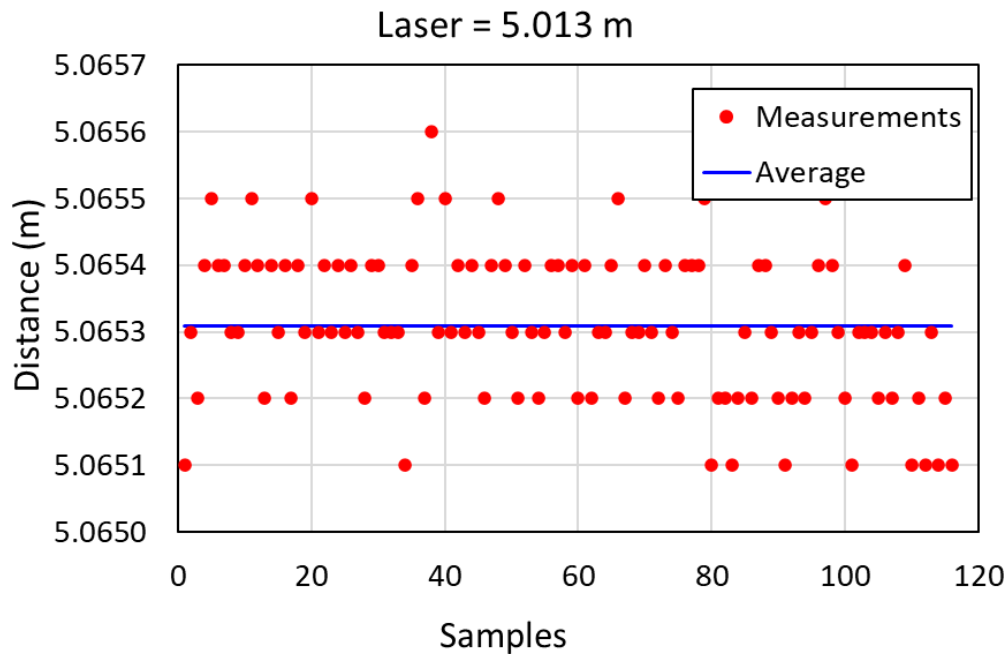
Test 4

Comparison between laser and radar, with the radar connected to a laptop. The scenario is outdoors, with the reflective target being a large flat concrete wall.



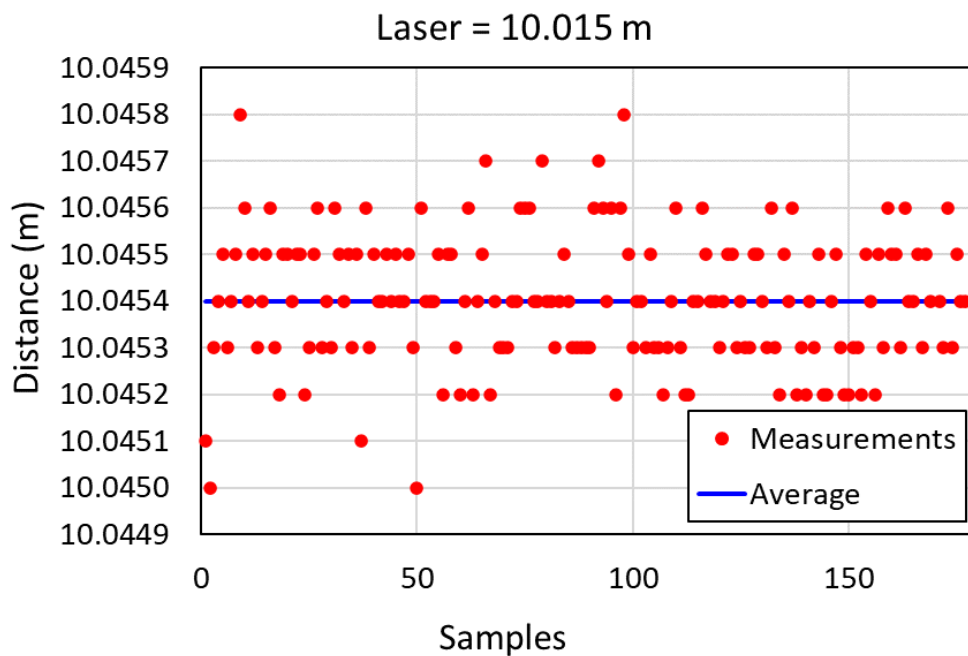
Configuration:

- Sample rate = 0.5 seg
- Maximum distance = 44 meters.



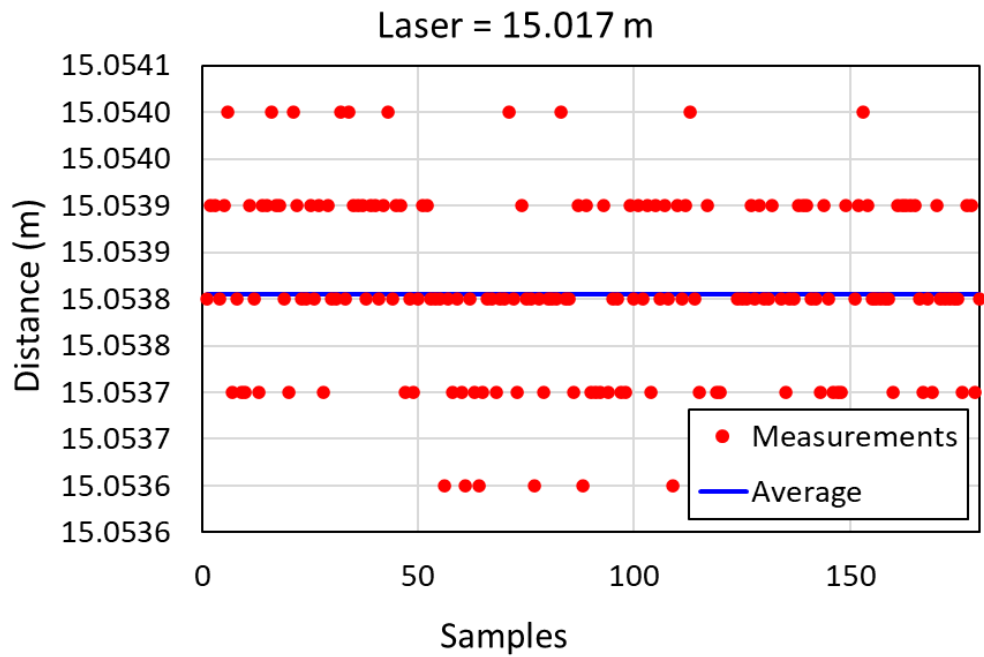
$$\text{Error} = 5.013 - 5.065 = -0.052 \text{ m}$$

$$\text{Deviation} = \pm 0.25 \text{ mm}$$



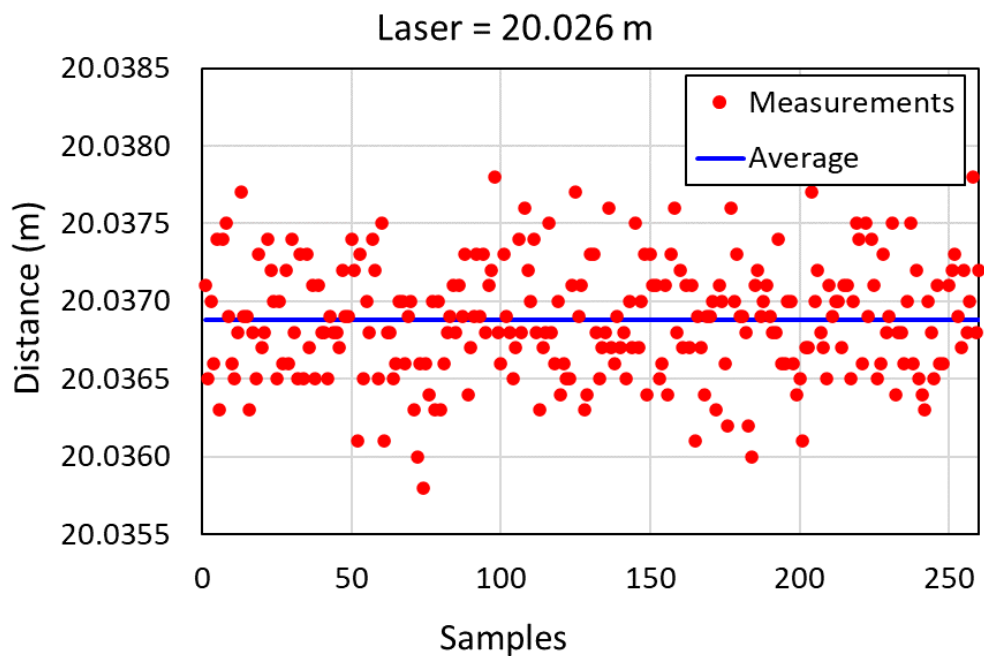
$$\text{Error} = 10.015 - 10.045 = -0.030 \text{ m}$$

$$\text{Deviation} = \pm 0.4 \text{ mm}$$



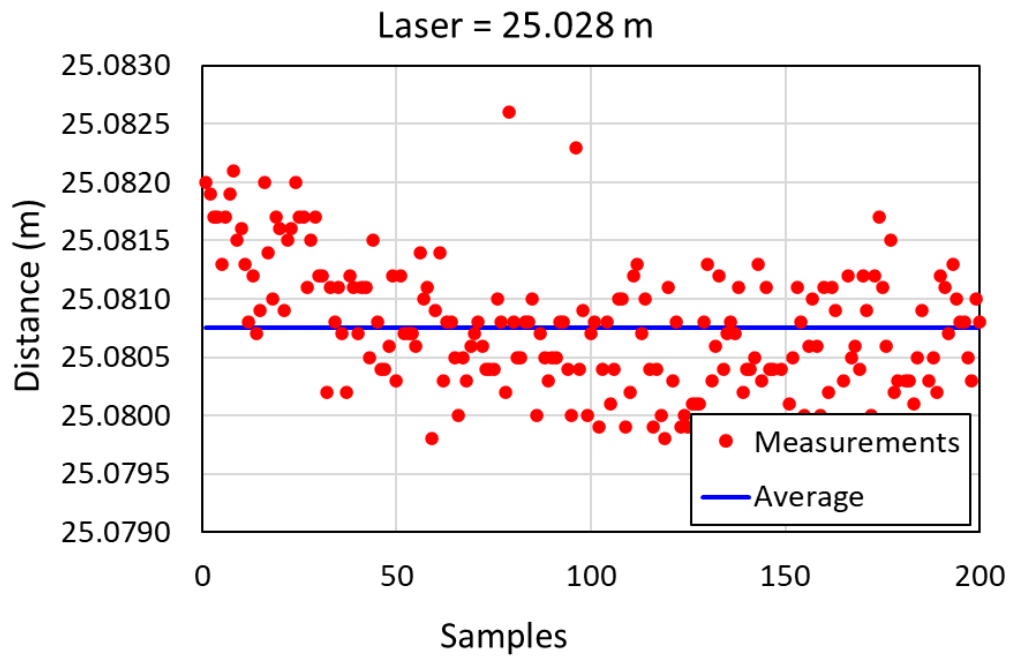
$$\text{Error} = 15.017 - 15.054 = -0.037 \text{ m}$$

$$\text{Deviation} = \pm 0.2 \text{ mm}$$



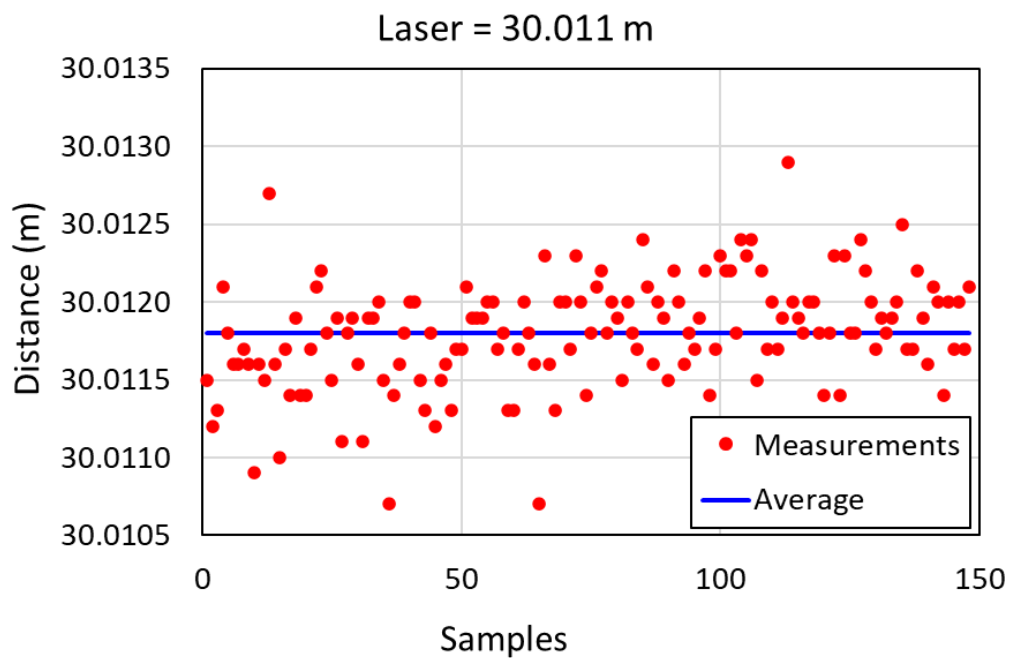
$$\text{Error} = 20.026 - 20.037 = -0.011 \text{ m}$$

$$\text{Deviation} = \pm 1 \text{ mm}$$



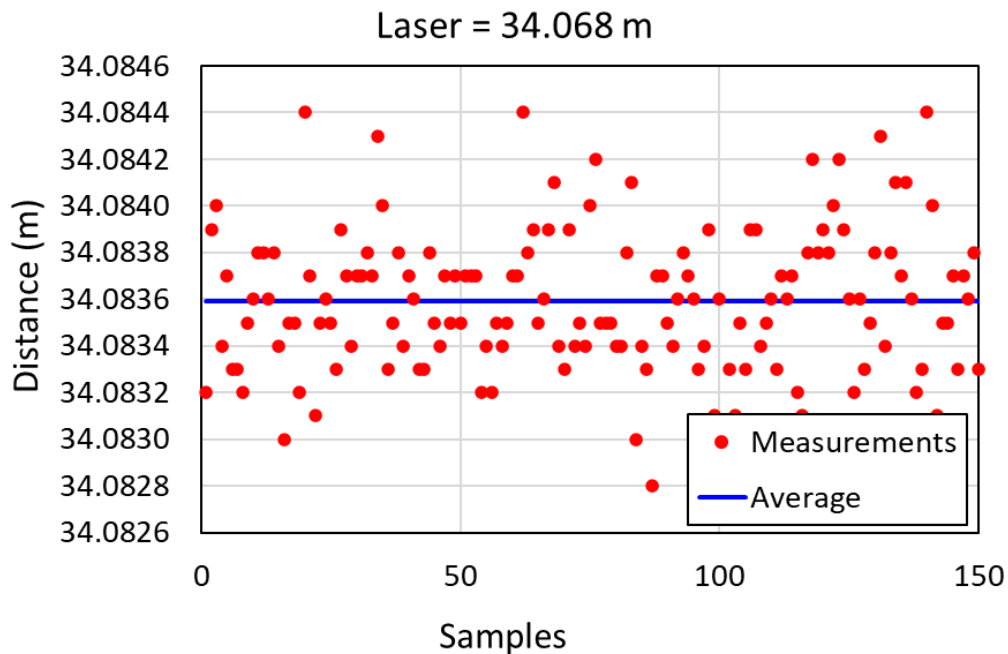
$$\text{Error} = 25.028 - 25.081 = -0.053 \text{ m}$$

$$\text{Deviation} = \pm 1.5 \text{ mm}$$



$$\text{Error} = 30.011 - 30.012 = -0.001 \text{ m}$$

$$\text{Deviation} = \pm 1.25 \text{ mm}$$



$$\text{Error} = 34.068 - 34.084 = -0.016 \text{ m}$$

$$\text{Deviation} = \pm 0.8 \text{ mm}$$

Conclusions:

- In an outdoor scenario where no other objects or movements influence the measurement, there is a very low deviation in the measurement. When the radar is close to the wall, it only detects the wall itself, resulting in a deviation below 1 mm. However, as the radar moves farther away, the left wall and the floor can influence the measurement because, although the beam is focused, a portion of the energy interacts with these surfaces.
- The error does not depend on the measured distance.
- During the test, we observed that when the radar was closer to the left wall, the distance detected in the first position (of the three available) was not the front wall; instead, it was, for example, the small corner present in the left window. The front wall was detected in the second position.